

Insider Trading Project - Data analysis in R

1. Project Overview

Goal:

Analyze U.S. politician insider-trading data to identify patterns, summarize key statistics, explore correlations, and conduct simple statistical inference. Present results with clear visualizations.

Data Source:

Processed dataset:

https://github.com/businessshark2061/Data-Analysis-in-R---Business-Sharks-group/blob/main/processed_data.csv

Dataset variables:

- `ticker`
- `tickertype`
- `company`
- `traded` (transaction date)
- `transaction_type` (buy/sell/exchange)
- `trade_size`
- `subholding`
- `description`
- `name` (politician)
- `filed` (filing date)
- `party`
- `district`

- `chamber` (House/Senate)
 - `excess_return`
 - `state`
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2. Derived Variables (to create)

- **Trade delay** = filed – traded
 - **Net trade direction**: buy = +1, sell = –1
 - **Dollar midpoint of trade size range**
 - **Politician-level activity metrics**: total trades, average trade size
 - **Company-level metrics**: number of trades per ticker
 - **Trade month and year** (for time trends)
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3. Correlations / Relationships to Explore

- **Chamber ↔ State** (relationship patterns)
 - **Excess return ↔ Chamber**
 - **Ticker type ↔ Excess return**
 - **Company ↔ Excess return** (best/worst performing stocks for politicians)
 - **Time effects**:
 - Does trade date or year affect excess return?
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4. Descriptive Statistics

3.1 Basic Descriptive Summaries

- Number of trades
 - Number of politicians
 - Number of companies
 - Most active politicians
 - Distribution of trade types (buy/sell)
 - Distribution of trade delay
 - Mean/median trade size
 - Top-traded tickers
 - Trades by month and year
 - Which specific day had the **most trades** (entire dataset)
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3.2 Group-Level Summaries

- Trades by party (D vs R)
 - Trades by chamber (House vs Senate)
 - Trades by state
 - Trades per politician
 - Trades by gender (if gender added manually)
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3.3 Excess Return Statistics

- Mean and median excess return
- Excess return by:
 - transaction type

- party
 - chamber
 - trade size category
 - company
 - Identification of:
 - Companies with the **most trades**
 - Companies where politicians had **positive vs negative excess return performance**
 - Top 10 most-traded companies vs the market benchmark
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Excess Return Interpretation Rules

If Purchase (Buy):

- Excess return $> 0 \rightarrow$ **Good trade**
- Excess return $< 0 \rightarrow$ **Bad trade**

If Sale:

- Excess return $< 0 \rightarrow$ **Good trade**
 - Excess return $> 0 \rightarrow$ **Bad trade**
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5. Statistical Inference

Confidence intervals

- CI for mean excess return
- CI for proportion of buy vs sell

Hypothesis tests

- Test if mean excess return > 0
- Do Democrats and Republicans differ in trade size?
- House vs Senate difference in filing delay?

(Use simple: t-tests or non-parametric equivalents.)

6. Visualizations

Basic Plots

- Histogram of trade sizes
- Histogram/density of excess return
- Boxplot of excess return by transaction type
- Bar plots:
 - trades by party
 - trades by chamber
 - trades by state

Relationship Plots

- Scatterplot: trade size vs excess return
- Time-series plot: number of trades over time

Bonus Idea

Map of the United States

- Total trades per state
- Total trade size per state
- Senate vs House trades by state